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CONCEPT MAP

Classification of Elements

MODERN PERIODIC TABLE

By Mosley-(1912)
The physical and chemical properties of elements are periodic functions of their atomic number

Periods and Groups

Groups:

Vertical columns contain elements having similar outer electronic configurations.

(18 Groups)

Periods:

Horizontal rows. Period number corresponds to the highest principal quantum number (n) of the elements.

Periods 1 to 6 contains 2,8,8,18,18,32 elements respectively.

Period 6 contains 14 elements (lanthanides)

Period 7 contains 14 elements (actinides)

Blocks

s- block:
group-1 : hydrogen and alkali metals
group 2 : alkaline earth metals

P block:
Group 13
Group 14
Group 15 :Pnictogens
Group-16 :Chalcogens
Group-17 : Halogens
Group 18: Noble gases

d block:

Contains 4 series

F block:
Contains 2 series
I series : lanthanides
II Series : actinides

PERIODIC TRENDS IN PROPERTIES OF ELEMENTS

TRENDS IN PHYSICAL PROPERTIES

Atomic Radii & Ionic Radii
- Along period decreases
- Within a group increases

Ionisation Enthalpy,

Electron affinity,

Electronegativity
- Along period increases
- Within a group decreases

TRENDS IN CHEMICAL PROPERTIES

- Valence or oxidation state
- Strength of oxyacids of non-metal
- Acidity of hydrides
- Acidity of oxides

Anomalous behavior of elements of II period: diagonal relationship : some elements of II period show similarities with diagonally placed III period elements

Eg : Li & Mg , Be & Al

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